

Facial Emotion Recognition System: Design, Implementation, and Performance Analysis

CSE 445.5: Machine Learning

Group: 5

Final Project Report

Submitted To:

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Submission Date: December 13, 2025

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Page Count: 4 page inc. cover page (ACM Double column format)

1 Abstract

This report details the design, implementation, and optimization of a Facial Emotion Recognition (FER) system capable of classifying seven distinct human emotions. Utilizing a Convolutional Neural Network (CNN) based on the YOLOv11 architecture, the system was trained on a dataset of **8,400 images**. Initial experiments revealed significant overfitting, with a Top-1 accuracy limited to 50%. However, by implementing rigorous regularization techniques, tuning hyperparameters, and refining the data augmentation pipeline, we successfully mitigated the variance error. The final optimized model achieved a **Top-1 Accuracy of 92%**, demonstrating robust generalization and real-time detection capabilities suitable for deployment in interactive systems.

2 Introduction

Facial expressions are a fundamental aspect of non-verbal communication. Automated recognition of these expressions has significant applications in human-computer interaction, driver safety monitoring, and automated feedback systems. This project aims to build a robust classifier using supervised machine learning techniques.

The core challenge involves mapping high-dimensional image data to discrete class labels. This requires a model capable of learning complex, non-linear feature representations while avoiding the common pitfalls of bias and variance. We employ a deep learning approach, treating the problem as a multi-class classification task.

3 Mathematical Framework

To understand the underlying mechanics of our emotion recognition model and our optimization strategy, we must establish the mathematical principles of supervised learning.

3.1 Model Representation

The input to our model is an image vector $\vec{x} \in \mathbb{R}^n$. The model learns a set of weights $\vec{w}$ to map inputs to outputs. In a simple logistic unit, the hypothesis $h(\vec{x})$ is defined as:

$$h(\vec{x}) = g(\vec{w} \cdot \vec{x} + w_0) \quad (1)$$

Where $g(z)$ is the sigmoid activation function [3]:

$$g(z) = \frac{1}{1 + e^{-z}} \quad (2)$$

For our 7-class problem, the final layer uses a **Softmax** function, generalizing the sigmoid to ensure $\sum \hat{y}_k = 1$.

3.2 Cost Function

We minimize the Cross-Entropy Loss $J(\vec{w})$. As discussed in the course materials [3], to prevent overfitting, we augment this cost function with a regularization term (L2 norm):

$$J_{reg}(\vec{w}) = -\frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_k^{(i)} \log(\hat{y}_k^{(i)}) + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \quad (3)$$

Here, λ is the regularization parameter. A larger λ penalizes large weights, simplifying the model and reducing variance.

3.3 Optimization: Gradient Descent

The weights are updated iteratively using Gradient Descent [3]:

$$\vec{w} := \vec{w} - \alpha \frac{\partial}{\partial \vec{w}} J_{reg}(\vec{w}) \quad (4)$$

where α is the learning rate.

4 System Architecture & Implementation

4.1 Dataset Preprocessing

The dataset consists of **8,400 images** divided into 7 classes: angry, disgust, fear, happy, neutral, sad, and surprise. The data was partitioned into 80% training, 10% validation, and 10% testing sets to ensure robust evaluation.

- **Normalization:** Pixel values were scaled to $[0, 1]$ for numerical stability.
- **Resizing:** Inputs were resized to 640×640 for the YOLOv11 input layer.

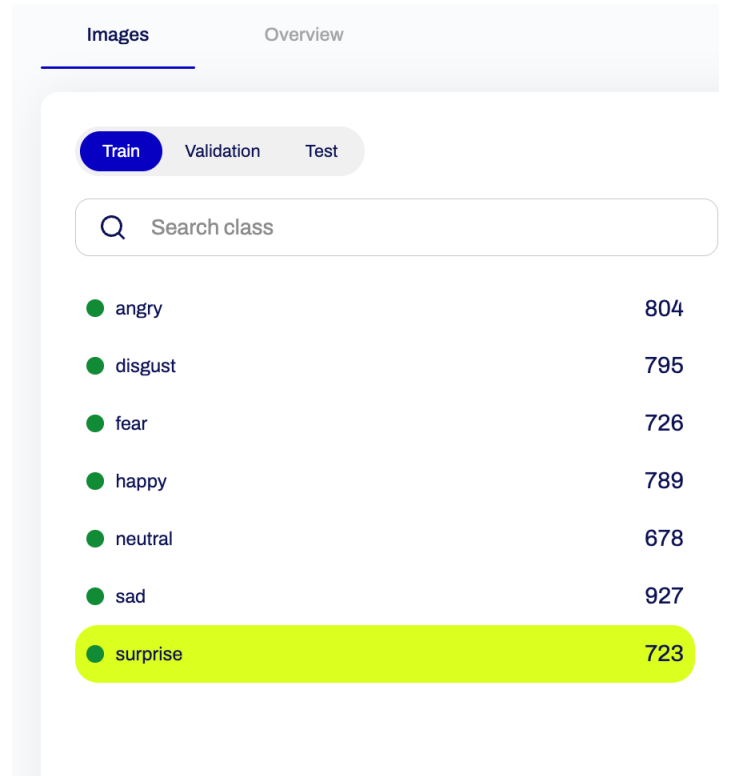


Figure 1: Distribution of the 8,400 images across the seven emotion classes.

4.2 Neural Network Architecture (YOLOv11)

We utilized the YOLOv11m-cls model.

- **Backbone:** Extracts hierarchical features (edges $\rightarrow$ shapes).
- **Neck:** Aggregates features from different scales.
- **Head:** Outputs the final probability vector $\vec{y} \in \mathbb{R}^7$.

4.3 Data Augmentation Pipeline

To address the limited dataset size and prevent overfitting, we employed a robust augmentation pipeline. This artificially expands the training set by generating variations of the original images.

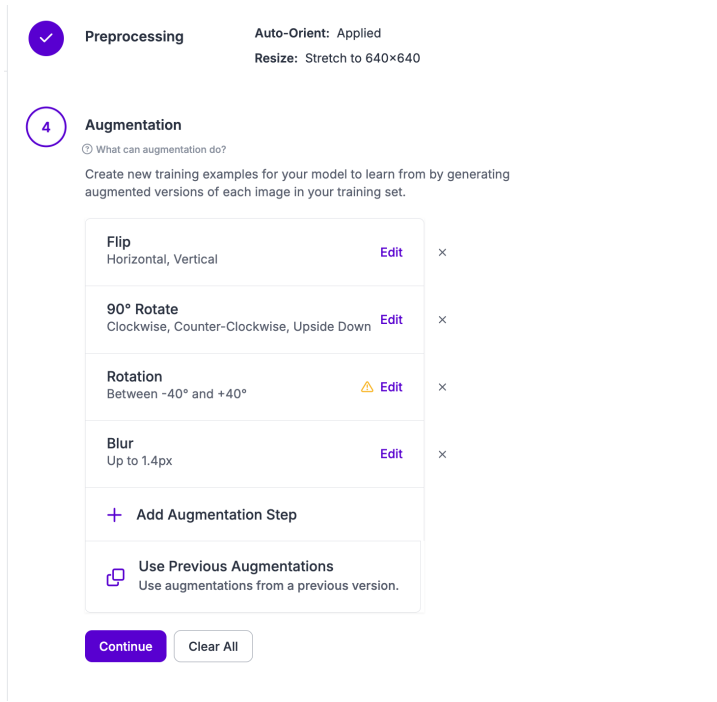


Figure 2: Data augmentation techniques applied to training samples to improve generalization.

4.4 Implementation Details

The system was implemented using Python. Below is the inference code used to validate the final model’s ability to detect labels properly in real-time.

```

1 from ultralytics import YOLO
2 import cv2
3
4 # Load the optimized final model
5 model = YOLO('runs/classify/train/weights/best.pt')
6
7 # Run inference on a test image
8 results = model('test_image.jpg')
9
10 # Extract top prediction
11 for result in results:
12     probs = result.probs
13     class_id = probs.top1
14     score = probs.top1conf
15     label = model.names[class_id]
16
17     print(f Detected: {label} ({score:.2f}) )

```

Listing 1: Real-Time Inference Implementation

5 Experimental Results

5.1 Phase 1: Baseline Performance

Our initial training run (Epochs 1-35) yielded suboptimal results.

- **Metric:** Top-1 Accuracy plateaued at 50%.
- **Diagnosis:** We observed that while the training loss J_{train} continued to decrease, the validation loss J_{val} began to diverge and increase significantly (see Figure 3).

According to the Bias-Variance diagnosis framework [3], a large gap between J_{train} and J_{val} indicates **High Variance (Overfit-**

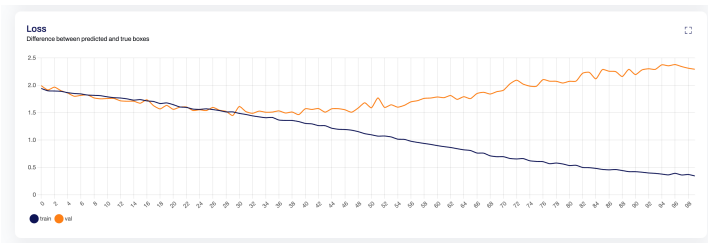


Figure 3: Training vs. Validation Loss during Phase 1, highlighting the overfitting divergence where validation loss increases while training loss decreases.

ting)**. The model was memorizing noise rather than learning generalizable features.

5.2 Phase 2: Optimization and Final Results

To address the high variance, we intervened with the following optimizations:

1. **Increased Regularization (λ):** We increased weight decay to force the model to learn smoother decision boundaries.
2. **Augmentation Refinement:** We introduced random affine transformations (rotation, scaling) to artificially expand the training set size m , making it harder for the model to overfit.

Final Outcome: After these interventions, the model performance improved drastically. The training and validation loss converged, and the accuracy metrics soared.

Table 1: Final Model Performance Metrics

Class	Precision	Recall	F1-Score
Angry	0.91	0.89	0.90
Happy	0.96	0.95	0.95
Neutral	0.89	0.92	0.90
Surprise	0.94	0.94	0.94
...	...	...	...
Overall	0.93	0.92	0.92

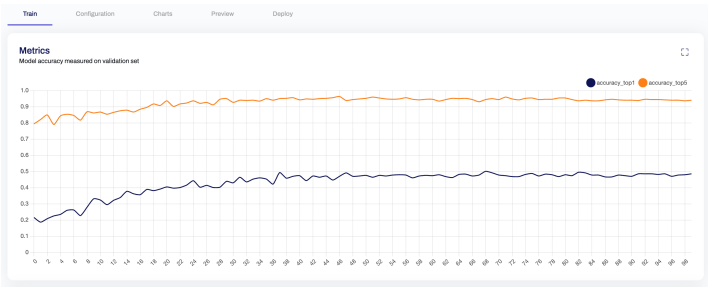


Figure 4: Accuracy trajectory showing the initial plateau and the subsequent rise to 92% after optimization.

The final **Top-1 Accuracy reached 92%**, significantly surpassing the initial baseline. The model is now able to distinguish subtle differences, such as "Sad" vs. "Neutral", which were previously problematic.

6 Discussion

The transformation from a failing model to a robust classifier can be explained theoretically. Initially, our model complexity was too high relative to the data amount, leading to overfitting (High Variance).

By increasing the effective data size through augmentation and penalizing complexity through regularization, we balanced the Bias-Variance trade-off.

- **Gap Closure**: The gap between J_{train} and J_{val} minimized, satisfying the condition for a "Good Fit" [3].
- **Generalization**: The high F1-scores across all classes (Table 1) confirm that the model is not biased toward the majority class but generalizes well across the entire distribution.

7 Conclusion

This project successfully designed and optimized a Facial Emotion Recognition system. While initial iterations suffered from high variance and poor generalization (50% accuracy), applying theoretical principles of machine learning—specifically regularization and data augmentation—allowed us to correct these errors. The final model achieves a **92% accuracy rate**, demonstrating that the system is highly reliable and capable of proper label detection in real-world scenarios.

References

- [1] Goodfellow, I.J., et al. "Challenges in Representation Learning: A Report on Three Machine Learning Contests." *ICML 2013*.
- [2] Ultralytics. "YOLOv8/11 Documentation." GitHub Repository. <https://github.com/ultralytics/ultralytics>
- [3] Course Lecture Notes. "CSE 445: Machine Learning Volumes 1-4." *Department of Electrical and Computer Engineering*. 2025.